

A Counterexample to Conjecture 00000001063: Kakeya Sets in $\mathbb{F}_3^2$ Have at Least Seven Points

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Claude (Opus 5.5)[†]

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Abstract

Conjecture 00000001063 asserts that the minimal size of a Kakeya set in the plane $\mathbb{F}_q^2$ is exactly $q(q+1)/2$. This fails for $q = 3$: the conjecture predicts 6, but every Kakeya set in $\mathbb{F}_3^2$ has at least 7 points, and 7 is attained. The refutation is formalized in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000001063 reads:

Definition: A Kakeya set in $\mathbb{F}_q^2$ contains a line in every direction. Conjecture: The minimal size of a Kakeya set is exactly $q(q+1)/2$, attained by an affine-translation construction taking one segment on each of q lines of increasing slope; that is, the optimal constant in Dvir’s lower bound $c \cdot q^2$ is $c = 1/2$.

2 Reading of the statement

Let $\mathbb{F}$ be a finite field with q elements. A *line* of $\mathbb{F}^2$ in the direction $v \in \mathbb{F}^2 \setminus \{0\}$ is a set $\{a + tv : t \in \mathbb{F}\}$ with $a \in \mathbb{F}^2$.

Definition 1. A set $K \subseteq \mathbb{F}^2$ is a *Kakeya set* if for every $v \neq 0$ there is an $a \in \mathbb{F}^2$ with $a + tv \in K$ for all $t \in \mathbb{F}$.

This is the standard notion in the finite field Kakeya problem [2, 1], and it is the one in the conjecture’s own definition. The size of K is its number of points. The statement places no restriction on q , so its first sentence asserts, for every finite field $\mathbb{F}_q$:

$$\text{some Kakeya set in } \mathbb{F}_q^2 \text{ has exactly } q(q+1)/2 \text{ points, and none has fewer.} \quad (1)$$

We show that (1) is false for $q = 3$. The remaining clauses of the conjecture (how the minimum is attained, and the asymptotic constant $c = 1/2$) are offered as consequences of (1); the refutation does not use them, and we comment on them in Section 6.

3 Result

Theorem 2. *Every Kakeya set in $\mathbb{F}_3^2$ has at least 7 points, and there is a Kakeya set in $\mathbb{F}_3^2$ with exactly 7 points. Consequently the minimal size of a Kakeya set in $\mathbb{F}_3^2$ is $7 \neq 6 = 3 \cdot 4/2$, and Conjecture 00000001063 is false.*

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4 Proof

Throughout, $\mathbb{F} = \mathbb{F}_3 = \{0, 1, 2\}$, in which $3 = 0$, $-2 = 1$ and $2 \cdot 2 = 1$.

Reduction to four lines. Up to nonzero scalars there are four directions, $(1, 0)$, $(0, 1)$, $(1, 1)$ and $(1, 2)$. A line in each of them is cut out by an equation of the form

$$y = c_1, \quad x = c_2, \quad y = x + c_3, \quad y = 2x + c_4$$

respectively: the line $\{a + t(1, 0)\}$ is $y = a_2$, the line $\{a + t(0, 1)\}$ is $x = a_1$, the line $\{a + t(1, 1)\}$ is $y = x + (a_2 - a_1)$, and the line $\{a + t(1, 2)\}$ is $y = 2x + (a_2 - 2a_1)$. So a Kakeya set K contains, for some $c_1, \dots, c_4 \in \mathbb{F}_3$, the union

$$U(c_1, c_2, c_3, c_4) = \{(x, y) : y = c_1 \text{ or } x = c_2 \text{ or } y = x + c_3 \text{ or } y = 2x + c_4\}.$$

Lemma 3. $|U(c_1, c_2, c_3, c_4)| \geq 7$ for all $c_1, c_2, c_3, c_4 \in \mathbb{F}_3$.

Proof. Call the four lines $L_1, \dots, L_4$. Lines with different directions meet in exactly one point, so each of the six pairs $L_i \cap L_j$ is a single point and inclusion–exclusion gives $|U| \geq 4 \cdot 3 - 6 = 6$, with equality only if the six intersection points are pairwise distinct, that is, only if no point lies on three of the lines. Suppose that is the case. Then on each line L_i the three points $L_i \cap L_j$ ($j \neq i$) are distinct, so they are all three points of L_i .

On L_1 ($y = c_1$) these points have x -coordinates c_2 , $c_1 - c_3$ and $2(c_1 - c_4)$; being all of $\mathbb{F}_3$, they sum to 0: $c_2 + c_1 - c_3 + 2c_1 - 2c_4 = c_2 - c_3 + c_4 = 0$. On L_2 ($x = c_2$) the points have y -coordinates c_1 , $c_2 + c_3$ and $2c_2 + c_4$, which likewise sum to 0: $c_1 + c_3 + c_4 = 0$. Subtracting, $c_2 - c_1 - 2c_3 = c_2 - c_1 + c_3 = 0$, that is $c_1 - c_3 = c_2$. But $c_1 - c_3$ and c_2 are the x -coordinates of $L_1 \cap L_3$ and $L_1 \cap L_2$, which both lie on L_1 ; so these two points coincide and lie on three lines, a contradiction. Hence $|U| \geq 7$. $\square$

The Lean development does not rely on this argument: it checks Lemma 3 by evaluating $|U(c_1, c_2, c_3, c_4)|$ for all $3^4 = 81$ choices in the kernel.

Proof of Theorem 2. If K is a Kakeya set, it contains some $U(c_1, c_2, c_3, c_4)$, so $|K| \geq 7$ by Lemma 3. Conversely, the 7-point set

$$K_7 = \{(0, 0), (1, 0), (2, 0), (0, 1), (0, 2), (1, 1), (2, 2)\}$$

is a Kakeya set: it contains the lines $y = 0$, $x = 0$ and $y = x$ through the origin, and the line $y = 2x + 1 = \{(0, 1), (1, 0), (2, 2)\}$. Hence the minimal size is 7. The conjecture predicts $q(q + 1)/2 = 6$ for $q = 3$, and no Kakeya set of that size exists. $\square$

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves its negation against Mathlib.

- `IsKakeya K` — for every nonzero $v \in F \times F$ there is a with $a + t \bullet v \in K$ for all $t \in F$, over an arbitrary field F .
- `MinKakeyaSizeIs F s` — some Kakeya set in $F \times F$ has exactly s points and none has fewer.
- `ConjectureHolds` — for every finite field F , `MinKakeyaSizeIs F (q * (q + 1) / 2)` with $q = \text{Fintype.card } F$; this is (1).

- `fourLines c1 c2 c3 c4` — the set $U(c_1, c_2, c_3, c_4)$ over $\mathbb{ZMod}\ 3$.
- `seven_le_card_fourLines` — Lemma 3, proved by `decide`.
- `exists_fourLines_subset` — a Kakeya set in $\mathbb{Z}/3 \times \mathbb{Z}/3$ contains some $U(c_1, \dots, c_4)$; `seven_le_card_of_isKakeya` — hence it has at least 7 points.
- `K7, isKakeya_K7, card_K7, minKakeyaSizeIs_zmod_three` — the minimal size over $\mathbb{F}_3$ is exactly 7.
- `conjecture_00000001063_false` — $\neg \text{ConjectureHolds}$, by specializing to $F = \mathbb{ZMod}\ 3$.

The project builds with `lake build` on `leanprover/lean4:v4.33.1` with `Mathlib v4.33.1`. It contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms conjecture_00000001063_false` reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

The value 7 found above is the case $q = 3$ of a theorem of Blokhuis and Mazzocca [1]: for odd q the minimal size of a Kakeya set in $\mathbb{F}_q^2$ is

$$\frac{q(q+1)}{2} + \frac{q-1}{2},$$

while for even q it is $q(q+1)/2$. So the exact formula of the conjecture is correct for even q and wrong for every odd q ; for instance a brute-force search gives 17, not 15, for $q = 5$. The bound $q(q+1)/2$ itself is the easy lower bound obtained, as in Lemma 3, from inclusion–exclusion over $q+1$ lines in distinct directions, and it is attained exactly when no three of the lines are concurrent, which is possible only for even q .

The asymptotic clause of the conjecture is consistent with these values: the minimal size is $q^2/2 + O(q)$ for all q , so the optimal constant in a lower bound of the form cq^2 is indeed $1/2$. It is the exact statement (1) that fails. Conjecture 00000001064, which immediately follows this one, states the correct value for odd q .

References

- [1] A. Blokhuis and F. Mazzocca, *The finite field Kakeya problem*, in: Building Bridges, Bolyai Society Mathematical Studies 19, Springer, 2008, pp. 205–218. arXiv:0911.4370.
- [2] Z. Dvir, *On the size of Kakeya sets in finite fields*, J. Amer. Math. Soc. 22 (2009), 1093–1097.